

{ Lab #1 :- Sampling & Handling the Sample }

* During Sampling, you take the following in Consideration

- 1- Data
- 2- Collection
 - A) representative Sample
e.g:- separate unit $\rightarrow$ sample must equal
 $\sqrt{\text{total sample}} \leq 10 \text{ units}$
 - B) Condition of the Sample
 - C) Aseptic condition

3- Handling (Holding).

4- Analysis:- All sample analysis as follow:-

a. Blending:-

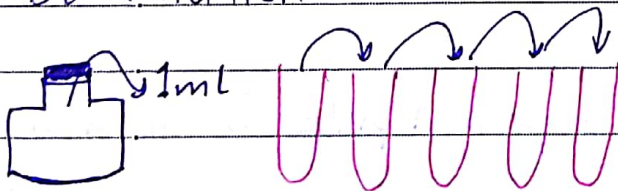
• If sample need blending:-

Weight sample $\rightarrow$ Steril blender $\xrightarrow{\text{add}}$ H_2O $\rightarrow$ blend

2 min low speed 2 min high speed

• If blending is not required, weigh sample is added to the dilution blank.

b. Dilution



c. Seeding:- Inoculation, use specific media.

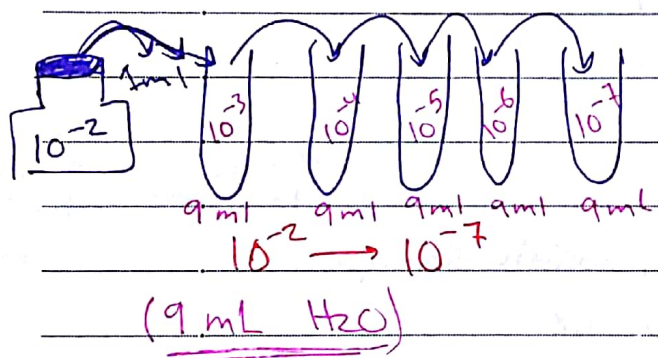
Lab Work:- In today lab, we will use fresh material e.g Tomato

* each 11g of fresh material = 1 g dry wt = 1 cm³
= 1 ml

Day 1:- Step (1)

Weight 11g of Tomato + 99 mL ~~St.D.H2O~~ → Steril ~~blend~~ blender
blend 2min, low sp. ←
" " , high " ←

Step (2) Dilution

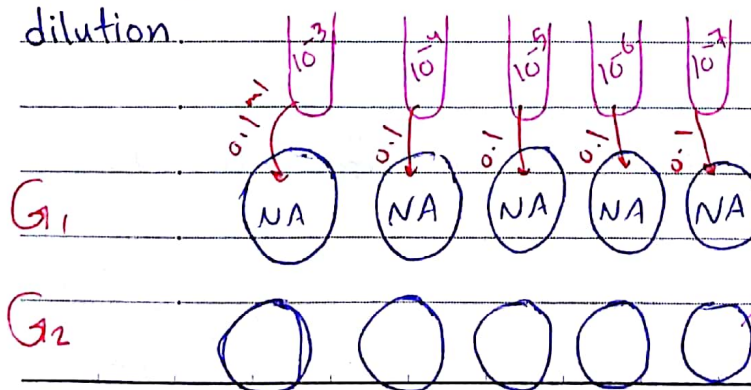


Homogenate
Sample (10⁻²)

work as
2 Groups

Step (3) Seeding

a) each group inoculate 0.1 ml over N.A plate from each dilution.



surface spreading method
L shape glass rod
↓
Incubation 24 hr / 35°C

Result:-

- * duplicate plate for each dilution.
- * be sure about number of Colony use colony counter.
- * Countable colonies:-

large colonies	3-20
medium "	3-300
Pin point "	3-1000

$$CFU = \frac{\text{\# of Average Colony} \times DF}{\text{inoculum}}$$

$$DF = \frac{1}{10^{-2}} = 10^2$$

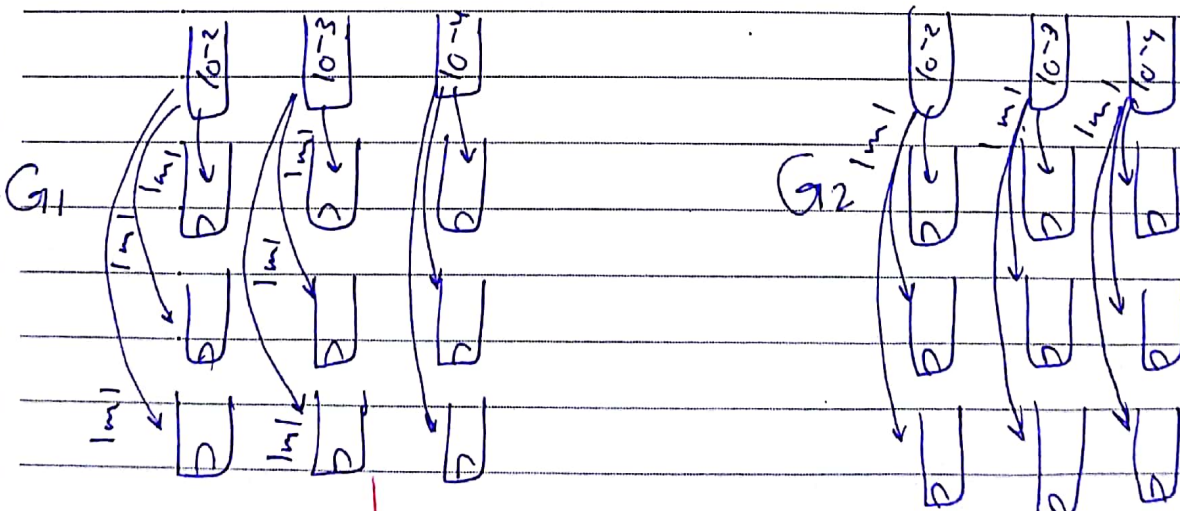
TNTC
Colonies

b) Check presence of Coliform and fecal Coliform

* Enterobacteriaceae.

* MPN $\Rightarrow$ determine \# of tubes that show gas after incubation time.

• For each dilution including $10^{-2} \rightarrow 10^{-7}$ take 3 test tubes of LST $\rightarrow$ Lauryl Sulfate Tryptose with Durham tube



Incubate (stand up)
24 hr / 35°C

* Check for gas production

* re incubation: 24 hr / 35°C

* Check for gas production

* ~~measure~~ MPN.
measure

DF	gas Production
10^{-2}	
10^{-3}	
10^{-4}	
10^{-5}	
10^{-6}	
10^{-7}	
10	

Lab #2:- Coliform & Fecal Coliform Organisms:-

Coliform:-

- belong to the Enterobacteriaceae.
- found in the intestinal tract of human and animals.
- used as good indicator for health hazard $\rightarrow$ water
- most are harmless.
- Mild heat treatment kill these organisms $\rightarrow$ e.g:- Milk Pasteurization.

Fecal Coliform:-

- Sub group of Coliform, e.g:- E. Coli.
- MPN $\rightarrow$ Method for Coliform and this method for drinking water.
 - $\rightarrow$ Determine the number of tubes that shows gas after incubation.
 - $\rightarrow$ 3-tests $\begin{cases} \text{Presumptive test} \\ \text{Confirm test} \\ \text{Complete test.} \end{cases}$

Lab Work:-

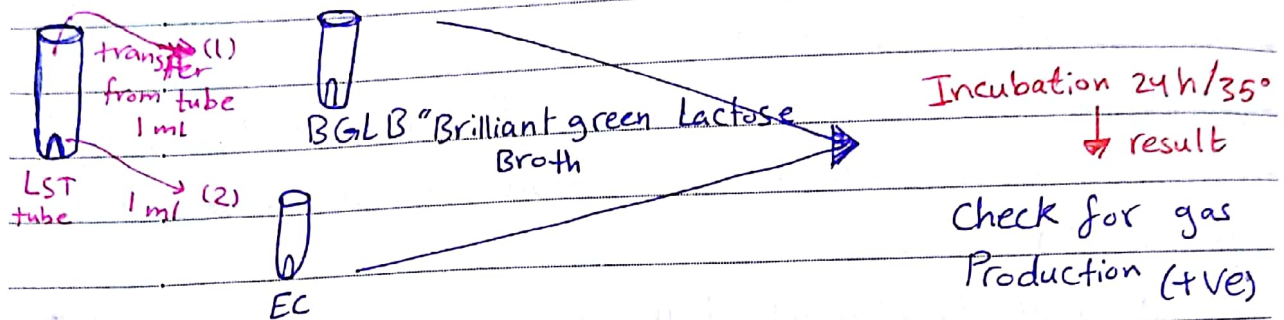
1] Presumptive Test:- (Day 1,2) $\Rightarrow$ Homogenous Sample "Tomato" (10^{-2})

Incubation of LST (Lauryl Sulfate Tryptose).

$\downarrow$ result

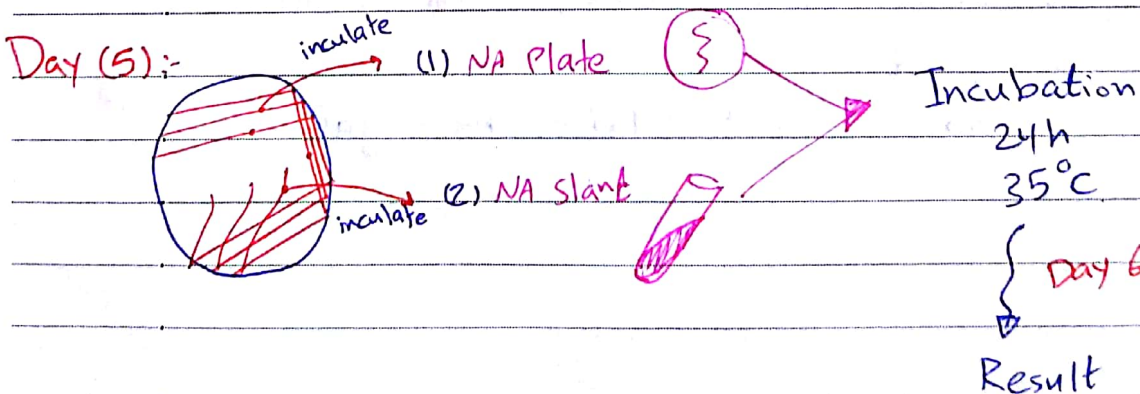
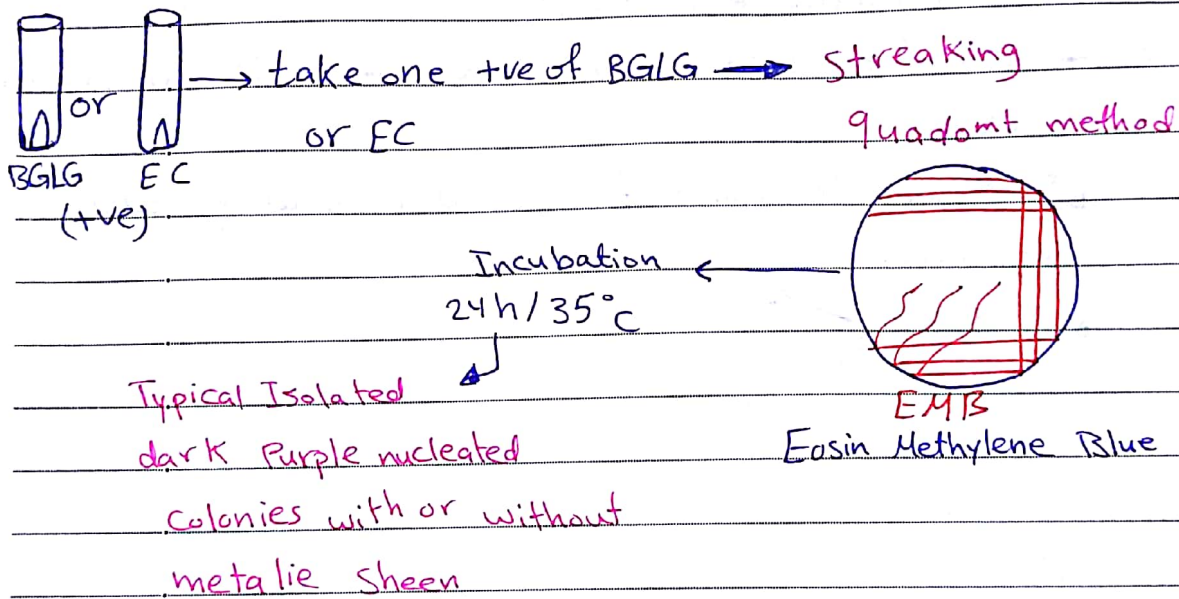
How many test tubes have gas.

[2] Confirm Test:- "Day 3"



[3] Complete Test:- "Day 4"

Check for gas production



Day 6. from NA Slant



1. Gram Staining :- gram -ve :- Cocco bacilli, non sporeformer,
sing single arrangement.

2. IMVIC test :-

a) SIM → take result after 24 hr
Sulfate Indol Method "add Kovacs reagent"

→ upper layer turns red +ve
→ Yellow -ve

b) MR-VP :- Methyl Red Vogas Proskaur → result 48h.

add methyl red reagent
(+ve) red
(-ve) yellow

add α -naphthol + 40% KOH

c) Simmon's Citrate → result 48h

+ve result indicated by growth in tube or blue color

3. Make pure culture

Report:

- * Count colonies
- * measure CFU/g of food "tabulated"
- * measure MPN
- * Result of IMVIC test "tabulated"
- * Gram Staining.
Gram -ve, coccobacilli, non sporeform, single arrangement.
- * Name of Bacteria
- * Pure culture.

Lab #3: Isolation of Salmonella

- Cause intestinal disease with vomiting and diarrhea.
- Isolate Salmonella from food, e.g.: Chicken skin.
- For Salmonella six groups include more than 1000 Serovars which identify according to:-

1. Somatic antigen (O)

2. flagellar antigen (H)

- 2 methods to isolate Salmonella:-

a. ELISA method

↳ e.g.: API-system or Enterotube

b. Traditional method:-

↳ (3) parts:- 1. Pre enrichment media → allow all types of microbes to grow, e.g.: lactose broth.

2. Selective media → allow Salmonella to grow suppress other, e.g.: Selenite broth.

3. Isolation then identification → XLD plate
"xylose Lysine di soxicholate agar"
↳ BS media
"Bismuth Sulfite agar plate"

as carbohydrate for bacteria

Lab Work:-

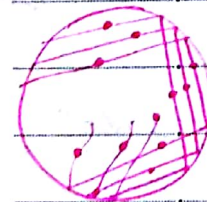
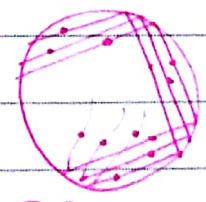
Day (1):-

take 10g of Chicken skin $\xrightarrow{\text{add it to}}$ lactose broth $\xrightarrow{\text{Shake by hand (10 min) (vigorously)}}$ $\xrightarrow{\text{Incubation 24 h } 35^{\circ}\text{C}}$ result $\rightarrow$ turbidity

Day (2):-

lactose with Chickenski $\xrightarrow{\text{Inoculate 1 ml}}$  $\xrightarrow{\text{Incubation 24 h / } 35^{\circ}\text{C}}$ result $\rightarrow$ Selenite transfer to pink color
turbidity Selenite broth (yellow)

Day (3):- Incultate from Selenite broth on:-

XLD BS
 
 $\xrightarrow{\text{Incubation 24 hr / } 35^{\circ}\text{C}}$ Result
 
XLD BS
"Red centered black colonies" "round nucleated black Jet colonies"

Report.:-

- * macro morphology for Salmonella over XLD, BS plate.
- * microscopic morphology for Salmonella "Gram Stain"
- * Write your interpretation.
- * pure culture.

Lab #4:- Isolation of Staphylococcus aureus

- * Isolation of staph. aureus from food $\rightarrow$ e.g.: falafel, meat.
- * Staphylococcus is:- G+ve, Cocci, grape-like arrangement (clusters),
* nonspore former.
- * produce enterotoxins:- A, B, C₁, C₂, D, E $\rightarrow$ Food poisoning.
- * make clot in blood plasma $\rightarrow$ Coagulase positive.
- * produce Nuclease enzyme $\rightarrow$ lysis the DNA.
- * Sensitive to lysostaphin enzyme $\rightarrow$ eg:- Staphylococcus staphylyticus
- * Also ~~find~~ found with micrococcus, staphylococcus epidermidis.

$\downarrow$
tetra

$\downarrow$
one of skin pathogenic
bacteria

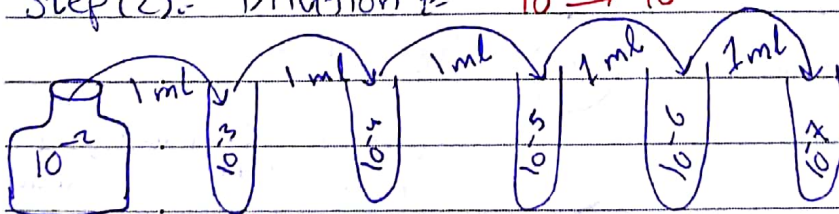
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Lab Work:-

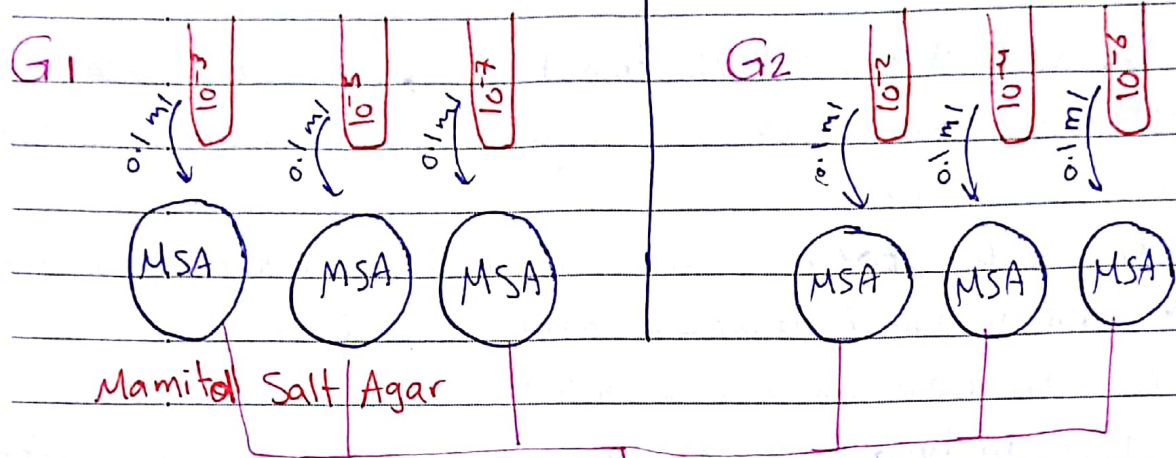
Day (1):- Step (1):- Blending:-

Food falafel + D.St.H₂O $\rightarrow$ Steril blender
 (11g) (99 ml)

blend $\rightarrow$ low speed 2min
 blend $\rightarrow$ high speed 2min
 $\downarrow$

Homognate Sample
(10⁻²)Step (2):- Dilution :- 10⁻² $\rightarrow$ 10⁻⁷

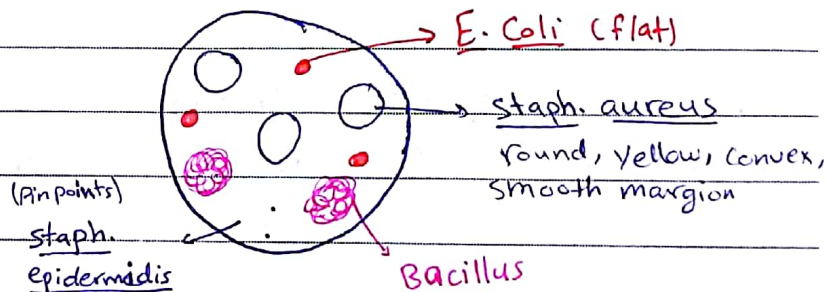
Step (3):- Seeding:-



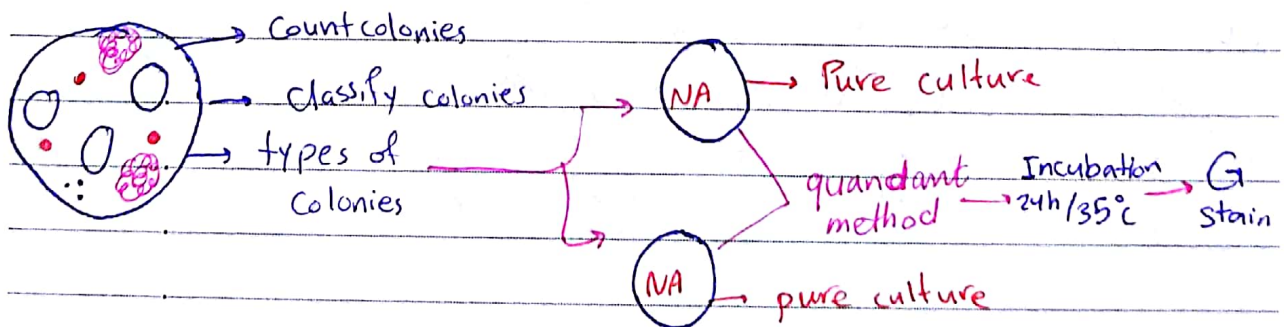
Surface spreading method
L-Shape glass rod

Incubation 48h

result

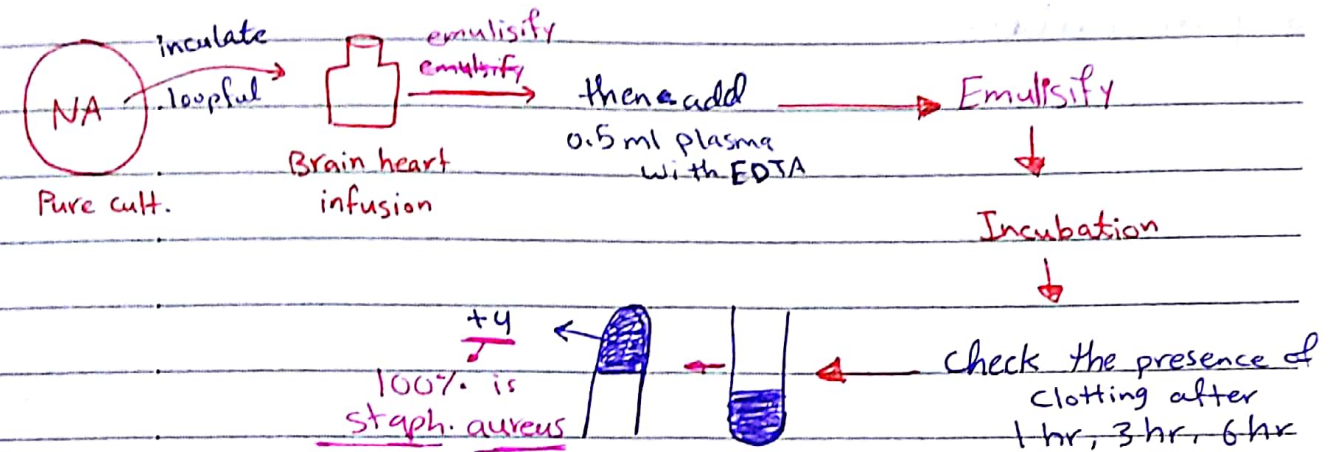


Day (2):-

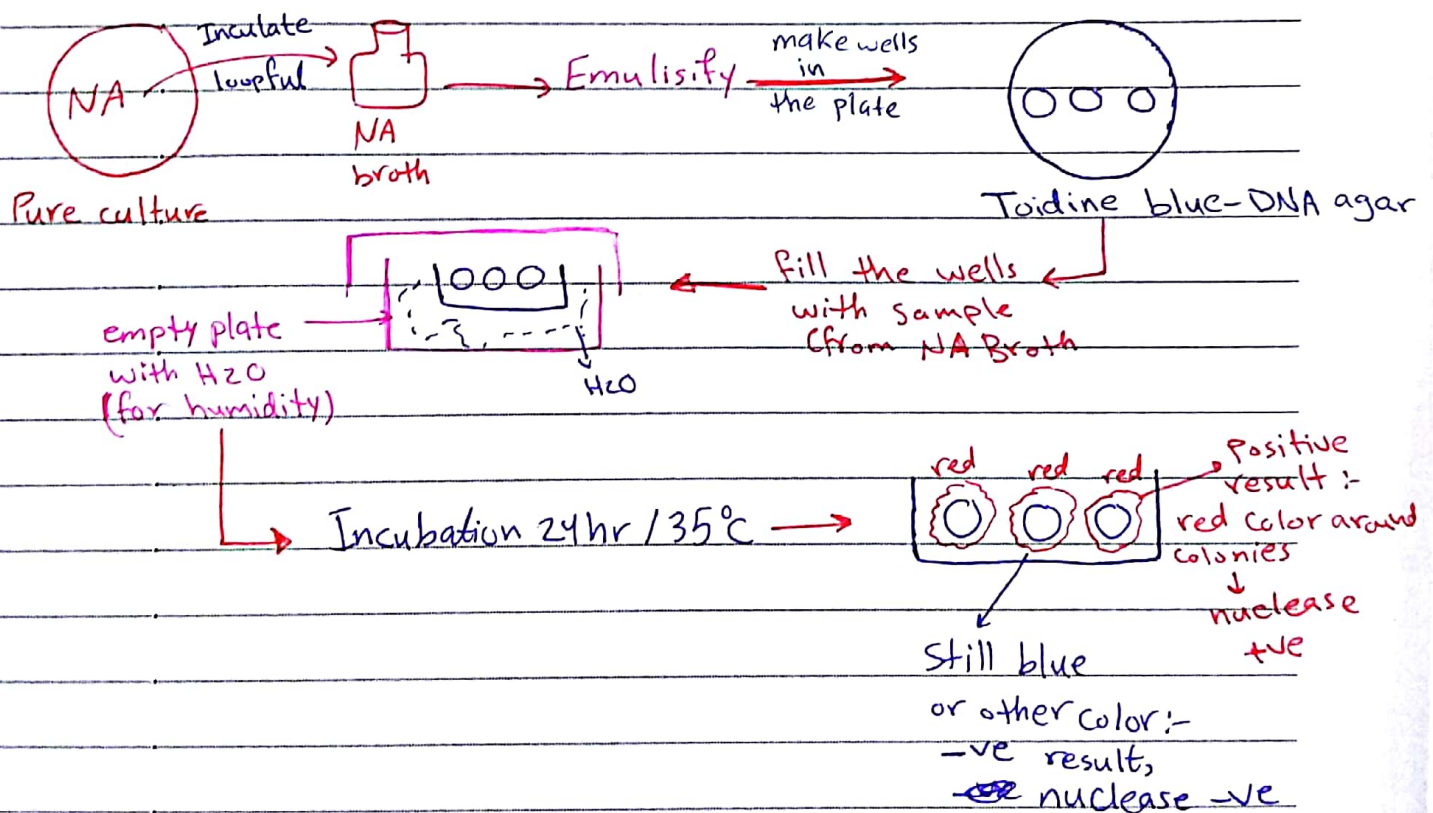


- Day (3):
- Gram stain
 - Coagulase test
 - Nuclease test

[b] Coagulase test:-



[c] Nuclease test:-

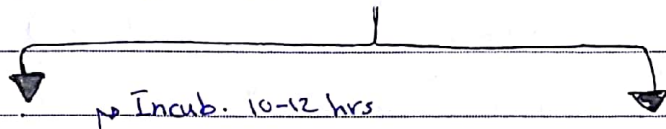


Reports

- * macroscopic morphology for Staph.
- * microscopic morphology for Staph. "Gram staining".
- * Result of coagulase and nuclease test "table"
- * Write your interpretation.
- * pure culture.

Lab #5:- Isolation of *Bacillus Cereus*

- * Gram +ve, rod, square end , central or subcentral endospore without swelling, arranged as chains.
- * Found in flour, vegetables, grains, soil, spices, - - -
- * produce 2 types of toxins:-



Thermolabile "heat labile"

Thermally stable

1-5 hrs

(break down by heat)

↓ called



or called

diarrhetic toxin

Vomiting or emetic toxin

↓ cause disease like

↓ cause disease like

*Clostridium Perfringens**Staphylococcus aureus*

this disease cause abdominal pain
with diarrhea.

↓ with

nausea and vomiting

↓ this strains isolated from



Proteinaceous food, pudding, sauce

rice

* * * * *

Lab Work:-

Day (1):- Step (1):-

1 g of soil + 99 ml of St. D. H₂O $\xrightarrow[\text{hand}]{\text{shake}}$ Homogenate Sample (10^2)

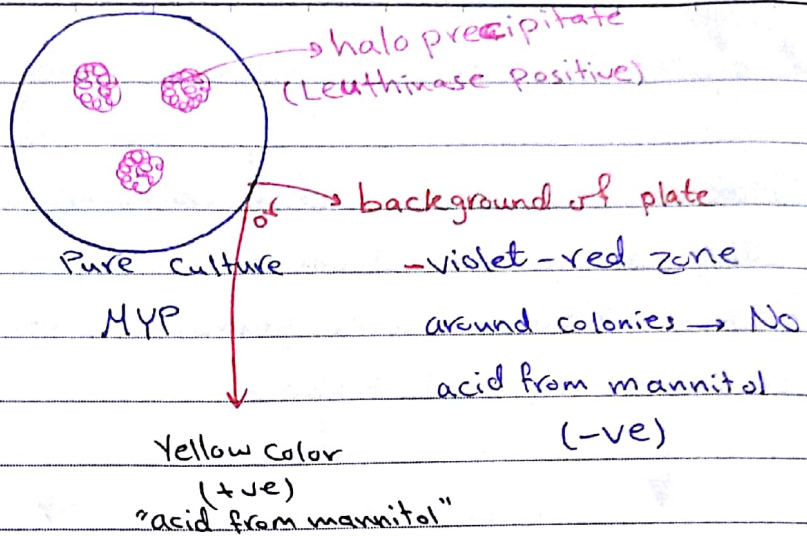
Step (2):- dilution $10^{-2} \rightarrow 10^{-5}$

Step (3):- Seeding:- for each group 2 plates of Phenol-red egg yolk
polymyxin agar (MYP) $\rightarrow$ Incubation 48 hr / 35°C

anti-fungus

selective media for *Bacillus Cereus*

Result:-



**** From pure culture :- make the following tests of Bacillus Cereus:-**

1] Phenol-red glucose broth with Durham tube :-

pH indicator

* acid $\rightarrow$ yellow *

neutral $\rightarrow$ red

base $\rightarrow$ pink

incubate by
loop

mix

incubation 48 hr/
35°C

result

Yellow with gas

لازم نشوف
اللون الأحمر
بعد 48
ساعة

✓ 100 Bacillus Cereus

mean (red $\rightarrow$ yellow)

2] Tyrosine agar Slant :-

Pure culture

Slant of
Tyrosine

incubate the surface
of the slant

incubation 48 hr / 35°C

result

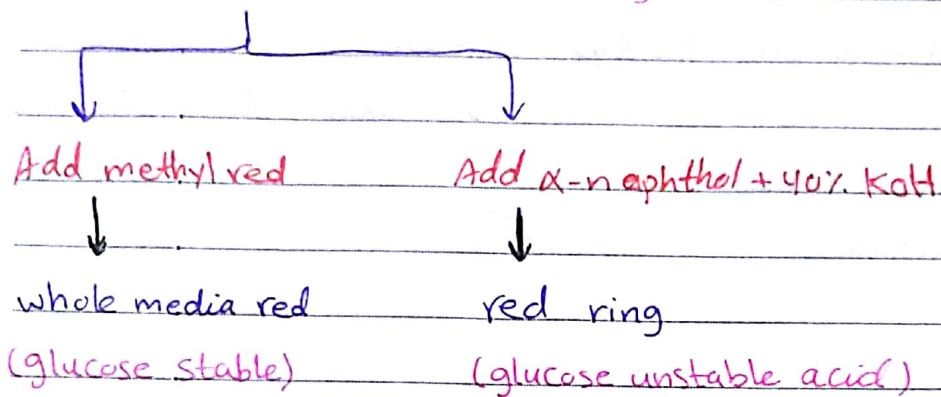
Colorless
area

mean

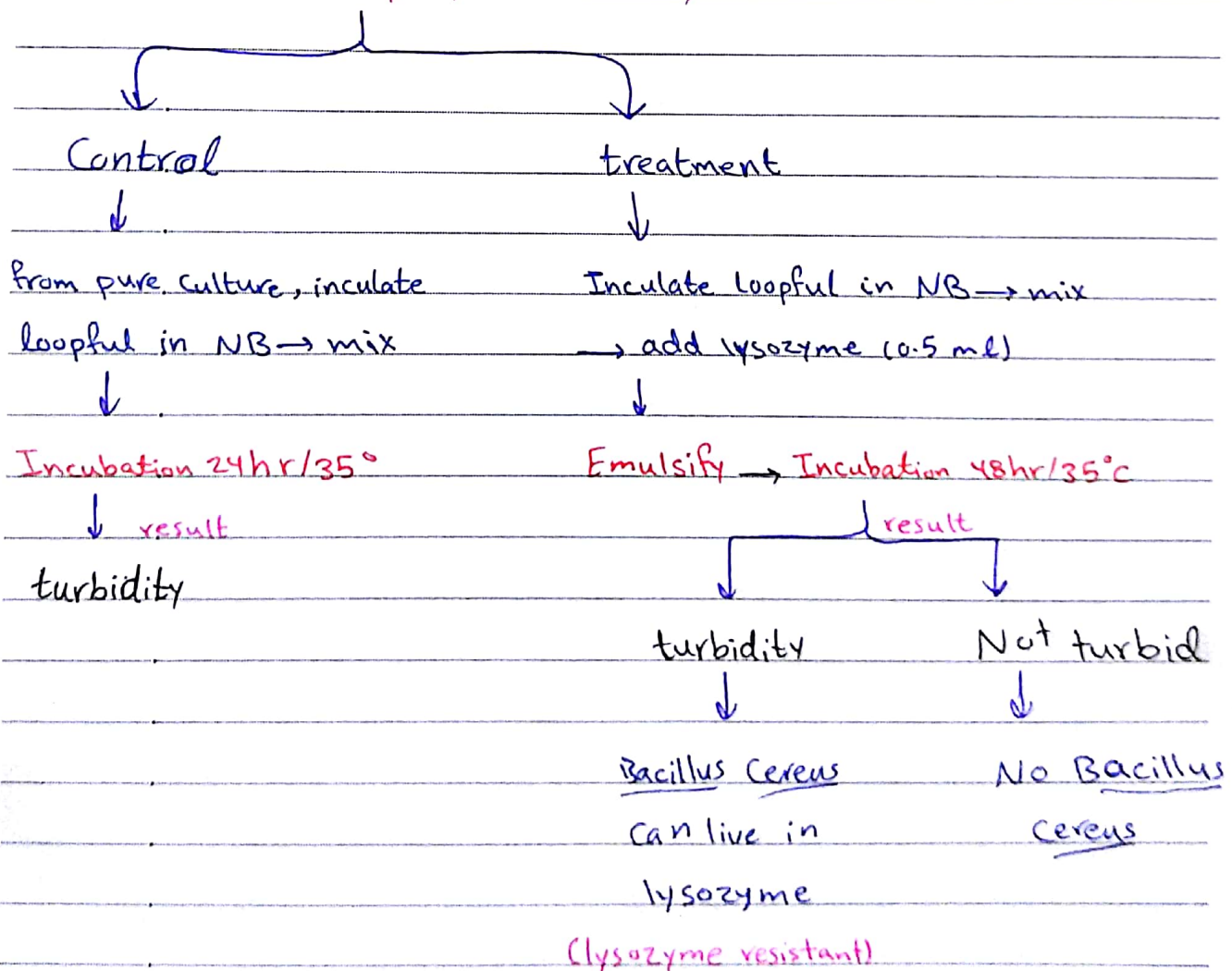
Bacillus / hydrolyze tyrosine.

Cereus

3 MR-VP: methyl-red Voges Proskauer test:



4 Growth with lysozyme (0.001%)



Report:-

- * Macroscopic Morphology for Bacillus Cereus.
- * Microscopic " " " " "Gram stain".
- * Results of 4 tests (table).
- * Write your interpretation.

Lab #6:- Carbohydrate:- starch hydrolysis

* Many microorganism produce enzymes ^{called} $\rightarrow$ hydrolases
 $\downarrow$ in which
 organic molecules $\xrightarrow[H_2O]{E}$ Smaller molecules

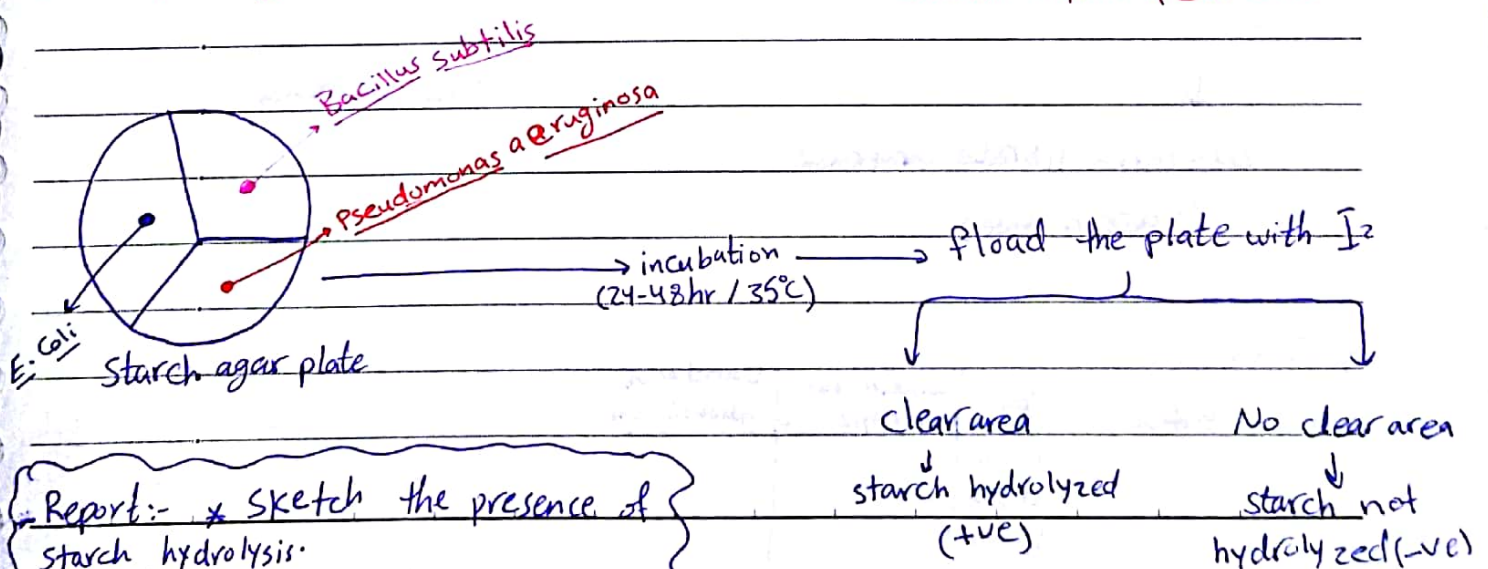
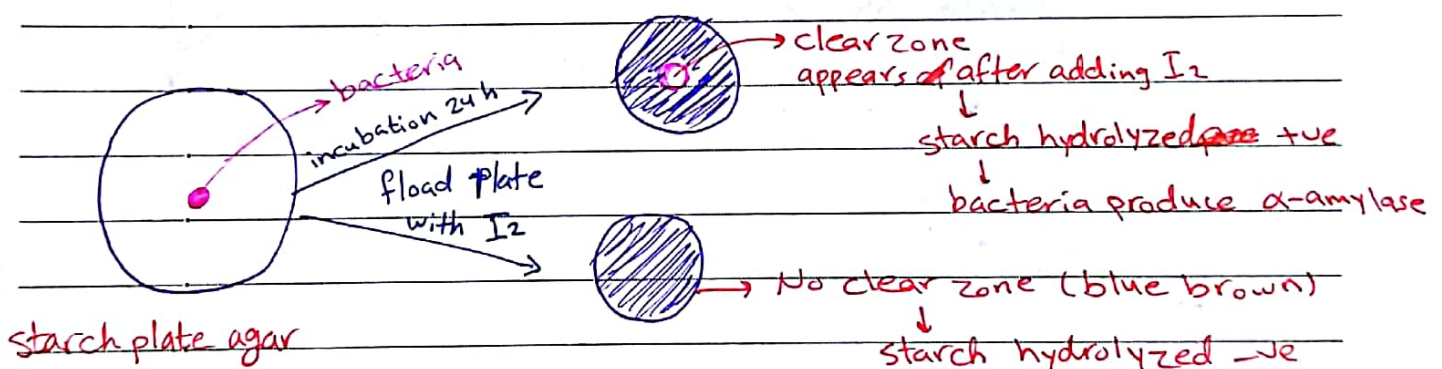
* Those microbial enzymes are important in industries (paper, dairy, baking, ...)

* Some microorganism ^{Produce} α -amylase.

* α -amylase hydrolyze starch ~~into~~ (amylase, amylopectin) $\rightarrow$ glucose, maltose, dextrins

Lab work :-

* Day (1) :- make a test to detect which M.O produce α -amylase :-



Report:- * Sketch the presence of starch hydrolysis.
 * Write your interpretation.

* Protein, Amino acid, and Enzymes :- Casein hydrolysis :-

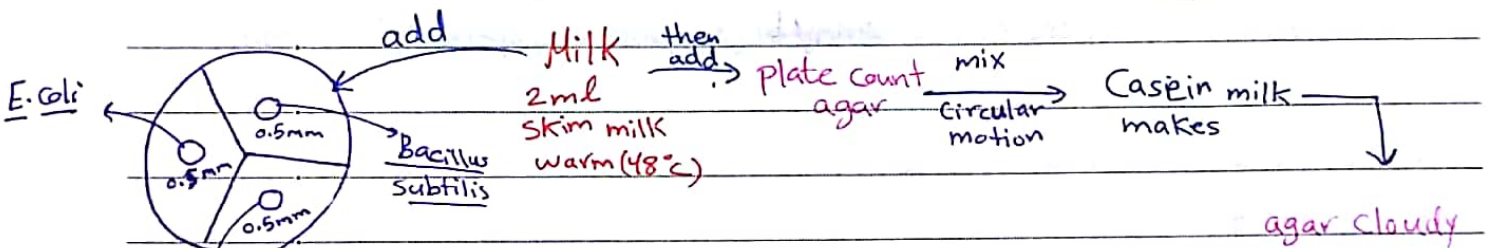
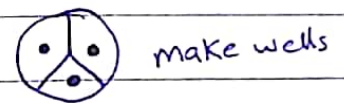
* Casein is large milk protein.

* Casein used by bacteria as carbon source.

* Some bacteria secret proteolytic enzyme Catalyze Casein Produce amino acid

Lab Work:

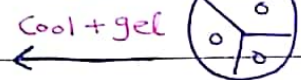
* mark the bottom of petri plate with 3 sections



Pseudomonas aeruginosa

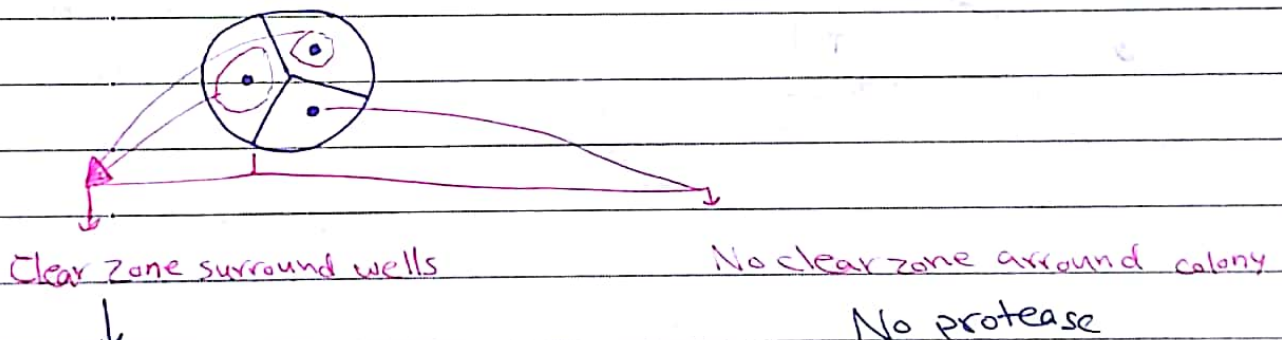
Fill wells with bacteria

make wells (plate)



skim milk agar plate

incubation (24-48 hr) / 35°

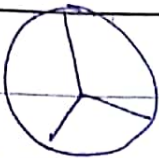


bacteria librate protease
(Casein +ve)

* Report:-

- make table

bac.	Casein +ve hydrolyse -ve	Diameter of hydrolysis zone
<u>E. coli</u>		
<u>B. sub</u>		
<u>P. aer</u>		

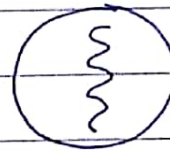
- draw  skim milk agar plate → show growth patterns

- Write your interpretation.

→ Isolate Lactobacillus:-

G + v, arrange as chain,
nonspore former.

NA Plate →



↓
Streaking by

Incubation 24h/35°C ← yoghurt

Lab #7: Quality of Milk

* Milk contaminated with bacteria → udder of cow
→ handling

* PAA → Pseudomonas → psychrotrophs
→ Acinetobacter → Gram -ve
→ Alcaligenes → Pathogenic (Human & Animal)
→ reduce quality of milk

* 3 methods to study PAA bacteria:-

[1] Standard Plate Count:-

→ Normal method = dilution, inoculate, -----

[2] Mosley method:-

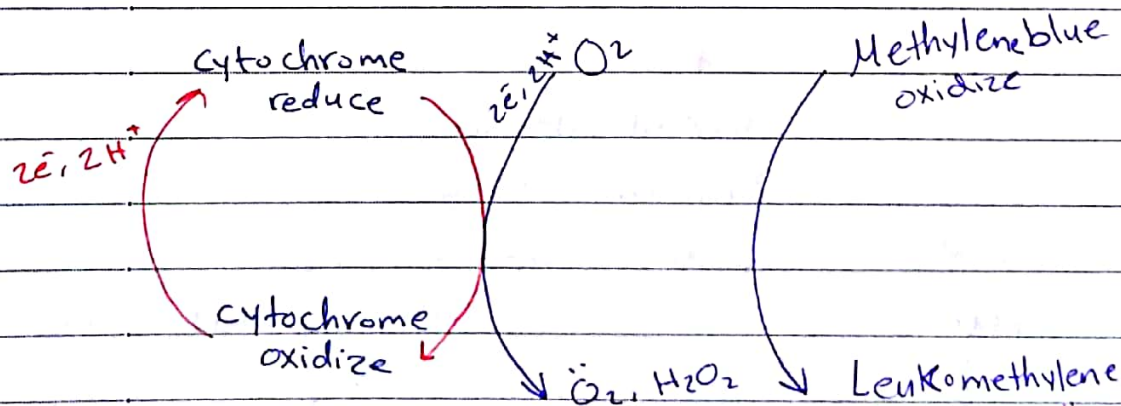
→ camper bpt - cooled & uncooled milk.

→ Psychrotrophic & thermotolerant bacteria.

[3] Reductase method:- (3)

→ through Electron-transport system

→ glycolysis & Krebs's cycle



* Milk "glucose" → degrade by PAA → NADPH → NADH → FADH → (36-38 ATP) ATP

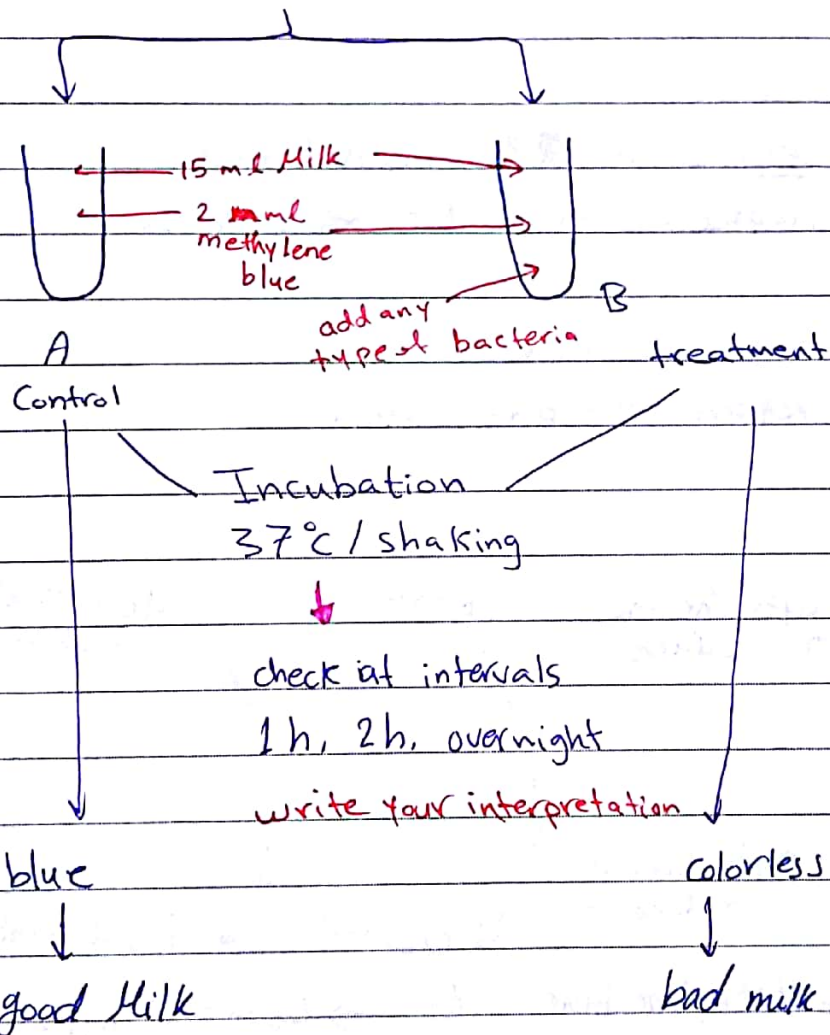
through glycolysis & Krebs cycle
form

* Absence of blue color indicate the presence of bacteria.

Lab Work:-

⇒ Quality of Milk:-

2 Samples of Milk:-



Yogurt Formation:-

* Yogurt is : granular, acidic, custard like texture, white app.

* Contains 3 types of bacteria:-

- Streptococcus thermophilus } - non pathogenic
- Lactobacillus acidophilus } - G +ve, large rod, nonspore former
- Lactobacillus bulgaricus } - arrange as chain.

* How can you make yogurt??

Milk (glucose) ^{those bacteria} convert to acid (coagulase protein of Milk).

* before pasteurization, 1ml of Milk must contain < 75000 cell/ml.

* after " " " " " " " < 15000 cell

⇒ Methods of Pasteurization:-

1) Normal pasteurized milks at $63^{\circ}\text{C} / 30 \text{ min}$.

2) Flash heating pasteurized " " " $72-75^{\circ}\text{C} / 15 \text{ sec}$

3) Ultra heating pasteurized " " " $121^{\circ}\text{C} / 1-2 \text{ sec}$

* * * * *

Lab Work :-

* Milk + teaspoon of yogurt mix flamed spatula Incubation $37^{\circ}\text{C} / 18\text{h}$ take results
% Write your interpretation

*  + Lactobacillus + 2 mL ST.D. H₂O $\downarrow$ Emulsify $\xrightarrow{\text{mix by hand}}$ Incubation 37°C / 18 hr $\rightarrow$ take results and write your interpretation

* Cheese Production :-

 + $\frac{1}{2}$ tablet of Renine + CaCl₂ $\xrightarrow{\text{mix}}$ Incubation $\rightarrow$ result.
"Calicum"
"Oxilate"

Lab #8 :- Honey as antimicrobial agent:-* Antimicrobial Antimicrobial properties of honey:-[1] osmotic effect :-

- honey is saturated solution of sugar (84%) a mixture of fructose and glucose.
- Water content $\approx 15-21\%$.
- free water molecule, (aw) water activity $\approx 0.56-0.62$
- bacterial growth inhibited if $aw < 0.94-0.99$

[2] Activity of honey

- honey is quite acidic $\rightarrow pH = 3.2-4.5$, optimum ~~temp~~ pH for microbes = $7.2-7.4$

[3] Hydrogen peroxide (H_2O_2)

- H_2O_2 is a major antimicrobial activity which produced enzymically in the honey.

Hypo Pharyngeal gland secret $\rightarrow$ glucose oxidase enzyme $\rightarrow$ nectar $\xrightarrow{\text{make}}$ Honey
(of Bee)

- H_2O_2 $\rightarrow$ effect as sterilizing agent $\xrightarrow{\text{only during}}$ ripening of honey
- H_2O_2 $\rightarrow$ enzyme inactive $\rightarrow$ full strength honey $\rightarrow$ has ~~negligible~~ negligible level of H_2O_2
because of presence of $\leftarrow$ metal ion and ascorbic acid

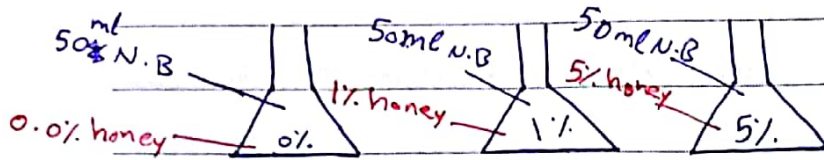
[4] Phytochemical factor:-

- In honey several chemicals with antimicrobial activity.

e.g.: ~~pinocebrin~~ pinocebrin, terpenes, benzyl alcohol, methyl syringate, syringic acid.

Lab work:-

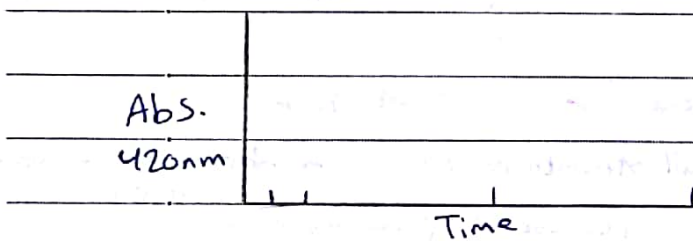
- * for each group, we need 3 flasks of nutrient broth with different conc. of honey:-



- N. Broth. 50 ml in each flask.

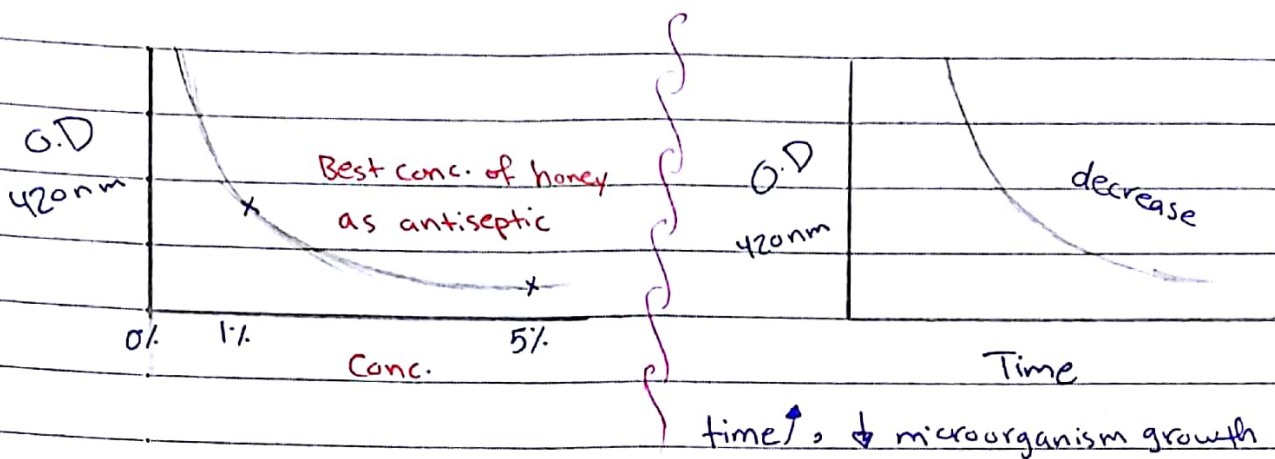
sterilization at 121°C for 10 min / 37°C (water bath).

- take blanks test tubes containing 0%, 1%, 5%.
- Inoculate 1 ml or 5 ml from any type of microbes E. coli or S. aureus (heavy growth).
- Inoculate in shaker water bath at 37°C .
- Take the reading every 1 hr, 2 hr then incubate in the usual incubator for 24 hr and also take the last reading after 48 hr. (absorbance at 420 nm).
- blot the curve:-



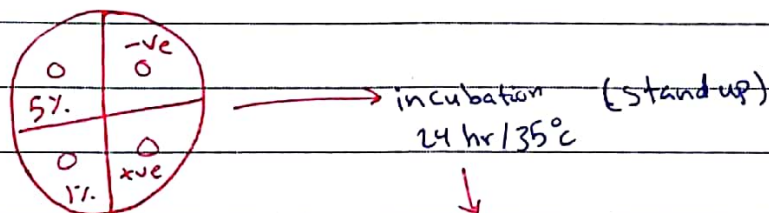
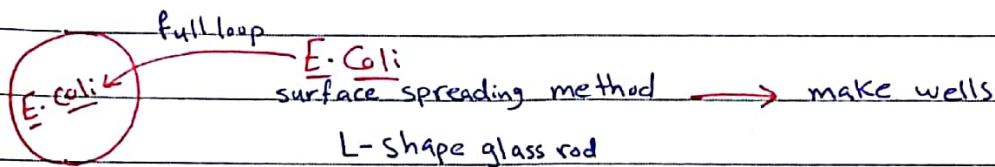
- tabulate the reading:-

Conc.	t_0	t_1	t_2	t_{24}	t_{48}
0%					
1%					
5%					



Effect of Clorex on the growth of bacteria:-

* prepare diff. conc. of clorex:- -ve control (H₂O), +ve control (100% clorex), 1% clorex, 5% clorex



measure the diameter of diffusion

Report:

- Draw sketch for (honey, clorex)
- Write your interpretation (both).

Lab #9. Microbial Spoilage of Pharmaceutical Products

* Spoilage of drugs due to
{

bacteria
fungi

* many drugs react with $O_2 \rightarrow$ oxidation $\xrightarrow{\text{cause}}$ change structure of compound.

* drugs exposed to light $\rightarrow$ photo degradation.

* * * * *

Lab Work :- How to test the presence of microorganisms in a drug sample??

Day (1):-

